

main.c



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Run

```
1 #include <stdio.h>
2 int main() {
3     int n, tq, i, time = 0, remain;
4     printf("Enter number of processes: ");
5     scanf("%d", &n);
6     int bt[n], rt[n], wt[n], tat[n];
7     printf("Enter Burst Time:\n");
8     for(i = 0; i < n; i++) {
9         scanf("%d", &bt[i]);
10        rt[i] = bt[i];
11    }
12    printf("Enter Time Quantum: ");
13    scanf("%d", &tq);
14    remain = n;
15    while(remain > 0) {
16        for(i = 0; i < n; i++) {
17            if(rt[i] > 0) {
18                if(rt[i] <= tq) {
19                    time += rt[i];
20                    wt[i] = time - bt[i];
21                    rt[i] = 0;
22                    remain--;
23                } else {
24                    rt[i] -= tq;
25                    time += tq;
26                }
27            }
28        }
29    }
30    for(i = 0; i < n; i++) {
31        tat[i] = wt[i] + bt[i];
32    }
33    printf("\nProcess\tBT\tWT\tTAT\n");
34    for(i = 0; i < n; i++) {
35        printf("%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);
36    }
37    return 0;
38 }
```

## Output

Enter number of processes: 7

Enter Burst Time:

5

7

4

4

5

7

1

Enter Time Quantum: 5

| Process | BT | WT | TAT |
|---------|----|----|-----|
| 1       | 5  | 0  | 5   |
| 2       | 7  | 24 | 31  |
| 3       | 4  | 10 | 14  |
| 4       | 4  | 14 | 18  |
| 5       | 5  | 18 | 23  |
| 6       | 7  | 26 | 33  |
| 7       | 1  | 28 | 29  |

=== Code Execution Successful ===